

CMT 442 – Neural Networks



Part 1: AI Ethics Foundations (45 Minutes)

- **Sources of bias in data and models:** Explain how historical inequalities get embedded into training data.
 - *Practical Example:* Discuss a localized scenario where a Kenyan fintech's mobile loan algorithm disproportionately rejects applicants from specific postal codes because the historical training data penalized areas with historically lower banking penetration.

- **Ethical risks in autonomous decision-making:** Explore what happens when human context is removed from critical choices.
 - *Practical Example:* Relate this to automated threat hunting or tax fraud detection systems (similar to a Tax360 project context), where a false positive could unjustly freeze a legitimate user's network access or financial accounts without recourse.
- **Societal impact of AI systems:** Discuss job displacement, algorithmic segregation, and erosion of privacy.

Part 2: AI Governance Frameworks (45 Minutes)

- **ISO/IEC 42001 management system:** Walk through this standard's plan-do-check-act (PDCA) cycle for AI. Highlight how it creates a verifiable AI accountability structure.

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- **NIST AI RMF governance function:** Explain the Map, Measure, Manage, and Govern core functions.
 - **Roles and accountability structures:** Detail the specific responsibilities of data scientists versus governance boards. Explain the role of AI oversight committees in enforcing policies.

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Part 3: Operationalizing Responsible AI (30 Minutes)

- **Human oversight mechanisms:** Discuss why human-in-the-loop controls are mandatory for high-risk systems.

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- **Model monitoring and review:** Discuss the concept of "model drift" where an AI's accuracy degrades over time as real-world data changes.
- **Incident management for AI failures:** Outline a step-by-step incident response plan for when an AI model causes harm despite meeting accuracy targets.

Module 6: AI Ethics, Governance & Responsible AI

Duration: 2 Hours

Standards Alignment: ISO/IEC 42001, NIST AI RMF – Govern

1. One-Page Infographic Summary (Textual Template)

- Bias & Fairness in AI Models
- Transparency & Accountability
- Human-in-the-Loop Controls
- AI Governance Structures
- Responsible AI Lifecycle

2. Detailed Lecture Notes

AI Ethics Foundations

- Sources of bias in data and models
- Ethical risks in autonomous decision-making
- Societal impact of AI systems

AI Governance Frameworks

- ISO/IEC 42001 management system
- NIST AI RMF governance function
- Roles and accountability structures

Operationalizing Responsible AI

- Human oversight mechanisms
- Model monitoring and review
- Incident management for AI failures

3. CAT-Style Scenario Questions

1. A facial recognition system shows racial bias. Identify two governance failures.
2. Explain how ISO/IEC 42001 addresses AI accountability.
3. Why is human oversight mandatory in high-risk AI systems?
4. Describe how bias audits should be conducted.
5. An AI model harms customers but meets accuracy targets. Explain the ethical issue.
6. How does governance differ from ethics in AI?
7. Explain the role of AI oversight committees.
8. Why is transparency critical for trust?
9. Describe lifecycle governance for AI systems.
10. Explain why responsible AI is a business risk issue.